3.13.5 Digital signal processing (A-level only)

3.13.5.1 Combinational logic (A-level only)

Content
Use of Boolean algebra related to truth tables and logic gates. $\overline{A}$ = not A $A \cdot B$ = A and B $A + B$ = A or B Identification and use of AND, NAND, OR, NOR, NOT and EOR gates in combination in logic circuits. Construction and deduction of a logic circuit from a truth table. The gates should be treated as building blocks. The internal structure or circuit of the gates is not required.

Counting circuits:

- Binary counter
- BCD counter
- Johnson counter.

Inputs to the circuits, clock, reset, up/down.

Outputs from the circuits.

Modulo- n counter from basic counter with the logic driving a reset pin.

The gates should be treated as building blocks. The internal structure or circuit of the gates is not required.

3.13.5.3 Astables (A-level only)

Content
The astable as an oscillator to provide a clock pulse.
Clock (pulse) rate (frequency), pulse width, period, duty cycle, mark-to-space ratio.
Variation of running frequency using an external RC network.
Knowledge of a particular circuit or a specific device (eg 555 chip) will not be required.